TERM PROJECT (Due date: June 4th, 2026, Thursday)

Objective

In this project, you will infer a Variable Structure Automaton (VSA) from observed interaction sequences generated in a P-model environment.
Your goal is to reconstruct:

- The state-action policy
- The state transition dynamics
- The environmental penalty structure
- The norm of behavior

This project is **data-driven and reproducible**. Your conclusions must be supported by **your dataset and your computations**.

System Description

The unknown system consists of:

- States: $\phi_1, \dots, \phi_5$
- Actions: $\alpha_1, \dots, \alpha_3$
- Environment: P-model (i.e., $\beta \in \{0,1\}$ to respectively represent favorable & unfavorable response)

Dataset

You are given a ***unique*** dataset of sequences:
Each entry: (sequence_id, step, $\Phi_{src}(n), \Phi_{dst}(n), \alpha(n), \beta(n)$)
Each sequence: $s_k = \langle t_{k,1}, \cdots, t_{k,J_k} \rangle$

Dataset Fingerprint

To validate your submission against this fingerprint, before modeling you need to compute and report

- Total number of rows
- Number of sequences
- Mean sequence length
- Empirical penalty rate per action
- A checksum

$$checksum = \sum_{all\ rows} step$$

<u>Tasks (1-6)</u>

Task 1 — Estimation

Estimate:

1. Policy: $\boldsymbol{P(\alpha_k|\phi_i)}$
2. Transition model: $\boldsymbol{P\left(\phi_j \middle| \phi_i, \alpha_k\right)}$
3. Penalty probabilities: $\boldsymbol{c_k = P(\beta = 1|\alpha_k)}$

<u>by computing and explicitly providing a table</u> $\boldsymbol{N(\phi_i, \alpha_k)}$, $\boldsymbol{N(\phi_i, \alpha_k, \phi_j)}$ and $\boldsymbol{N(\alpha_k, \beta)}$ **(raw counts; $\boldsymbol{N}$ stands for "number").**

Make the following consistency checks:

- $\sum_k \boldsymbol{P(\alpha_k|\phi_i)} = 1$
- $\sum_j \boldsymbol{P\left(\phi_j \middle| \phi_i, \alpha_k\right)} = 1$
- **Computed $\boldsymbol{c_k}$ and empirical frequencies match!**

Task 2 — Optimal Action Identification

Determine:

$$\boldsymbol{\alpha^* = \alpha_{argmin_k c_k}}$$

Task 3 — ε-Optimality Analysis

Estimate:

$$\boldsymbol{\varepsilon = 1 - min_i P(\alpha^*|\phi_i)}$$

Interpret your result:

- Is the system strongly optimal?
- Is it noisy?
- Is behavior consistent across states?

Task 4 — Simulation

Using your learned model:

- Generate synthetic sequences
- Compare with dataset:
  - action distribution
  - penalty rates
  - state visitation

To validate compute absolute probability difference for each! (i.e., $|\boldsymbol{P_{data} - P_{sim}}|$)

Task 5 — Perturbation Experiment

*Create a **modified dataset**:*

- *Remove **20% of transitions randomly***

Recompute:

- $\boldsymbol{c_k}$
- *policy*
- $\boldsymbol{\varepsilon}$

Analyze:

- Which quantities are stable?
- Which degrade most?
- Why?

Task 6 — Model assumption check
Evaluate:

Does the dataset support the P-model assumption?

Specifically:

- Compute c_k separately per state
- Compare across states

Deliverables

- **Report (8–12 pages)** Must follow this structure:
  - **Section 1 — Dataset fingerprint**
  - **Section 2 — Raw counts**
  - **Section 3 — Probability derivations**
  - **Section 4 — Optimality analysis**
  - **Section 5 — Simulation results**
  - **Section 6 — Perturbation analysis**
  - **Section 7 — Model critique**
  - **Code (well-documented)**
    - **Must reproduce all results**
    - **Must generate all tables in report**
    - **Must be runnable**
  - **Learned matrices/tables**
    - **Counts**
    - **Policy matrix**
    - **Transition matrix**
    - **Penalty estimates**

Evaluation

Component	**Weight**
Model correctness	20%
Policy estimation	15%
Transition modeling	15%
Penalty estimation	10%
ε-analysis	15%
Simulation & validation	10%
Perturbation analysis	10%
Insight & discussion	5%